Math 242, S13
Exam 1

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points) Solve the initial value problem

$$\frac{dy}{dx} = \frac{-3}{\sqrt{2x+1}}, \qquad y(4) = -1.$$

Solution: $y(x) = \int \frac{-3}{\sqrt{2x+1}}\, dx$

$= -3\sqrt{2x+1} + C$

$y(4) = -1 \Rightarrow -3\sqrt{2\cdot 4+1} + C = -1$

$-9 + C = -1$

$C = 8$

Thus, $y(x) = -3\sqrt{2x+1} + 8.$

Problem 2. (20 points) Find a general solution of the following differential equation:

$$\frac{dy}{dx} = 1 + x^2 + y^2 + x^2y^2.$$

(*Hint: Factor the right-hand side.*)

Solution: $\frac{dy}{dx} = 1+x^2+y^2(1+x^2)$

$= (1+x^2)(1+y^2)$ → seperable

$\Rightarrow \int \frac{1}{1+y^2}\, dy = \int (1+x^2)\, dx$

$\tan^{-1}(y) = x + \frac{x^3}{3} + C$

$y(x) = \tan\left(x + \frac{x^3}{3} + C\right)$

Problem 3. (20 points) Find the particular solution of the initial value problem:

$$(1+x^2)\frac{dy}{dx} - 2x(1+x^2)y = 2xe^{x^2}, \qquad y(0) = 2.$$

(*You need first transform the equation into the standard form of first-order linear differential equations.*)

Solution: standard form

$$\frac{dy}{dx} - 2x\,y = \frac{2xe^{x^2}}{1+x^2}$$

Then we have $P(x) = -2x$, $Q(x) = \frac{2xe^{x^2}}{1+x^2}$

Integrating factor:

$$\rho(x) = e^{\int P(x)dx} = e^{\int (-2x)dx} = e^{-x^2}$$

$$\Rightarrow e^{-x^2}\frac{dy}{dx} - 2x e^{-x^2} y = \frac{2x}{1+x^2} \cancel{e^{x^2}} \cancel{e^{-x^2}}$$

$$\frac{d}{dx}[e^{-x^2}y] = \frac{2x}{1+x^2}$$

$$e^{-x^2}y = \int \frac{2x}{1+x^2}\,dx$$

$$e^{-x^2}y = \ln(1+x^2) + C$$

$$y = e^{x^2}(\ln(1+x^2) + C)$$

$$y(0) = 2 \Rightarrow e^0(\ln 1 + C) = 2 \Rightarrow C = 2$$

Thus,

$$y(x) = e^{x^2}(\ln(1+x^2) + 2)$$

Problem 4. (20 points) A certain city had a population of 25000 initially and a population of 30000 after 10 years. Assume that its population grows at a constant rate k per year.

(a) Write down the differential equation for the population $P(t)$ of the city and find its solution.

Solution: $\frac{dP}{dt} = kP$

$\Rightarrow P(t) = Ce^{kt}$

(b) Determine all constants using the given conditions.

Solution: $P(0) = 25000 \Rightarrow Ce^{0} = 25000$

$\Rightarrow C = 25000$

$\Rightarrow P(t) = 25000\, e^{kt}$

$P(10) = 30000 \Rightarrow 25000 \cdot e^{10k} = 30000$

$e^{10k} = \frac{30000}{25000}$

$\Rightarrow k = \frac{1}{10} \ln\left(\frac{6}{5}\right) \approx 0.0182$

Thus $P(t) = 25000\, e^{0.0182t}$

(c) How long does it take for the population to reach 40000?

Solution: To solve $P(t) = 40000$

$25000\, e^{0.0182t} = 40000$

$e^{0.0182t} = \frac{40000}{25000} \frac{8}{5}$

$t = \frac{1}{0.0182} \ln\left(\frac{8}{5}\right)$

≈ 25.8 years

Problem 5. (20 points) Find a general solution for the following differential equation using the substitution method:

$$x^2y' = xy + y^2$$

(*Hint: Transform the equation into the form of homogeneous equation* $y' = F(\frac{y}{x})$ *and then use the substitution* $v = \dfrac{y}{x}$.)

Solution: $y' = \dfrac{xy+y^2}{x^2}$

$$\Rightarrow y' = \frac{y}{x} + \left(\frac{y}{x}\right)^2$$

Let $v = \frac{y}{x}$, $y = vx$, then $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

$$\Rightarrow \cancel{v} + x\frac{dv}{dx} = \cancel{v} + v^2$$

$$x\frac{dv}{dx} = v^2.$$

$$\int \frac{1}{v^2}\,dv = \int \frac{1}{x}\,dx$$

$$-\frac{1}{v} = \ln|x| + C$$

$$-\frac{x}{y} = \ln|x| + C$$

$$y(x) = \frac{-x}{\ln|x| + C}$$